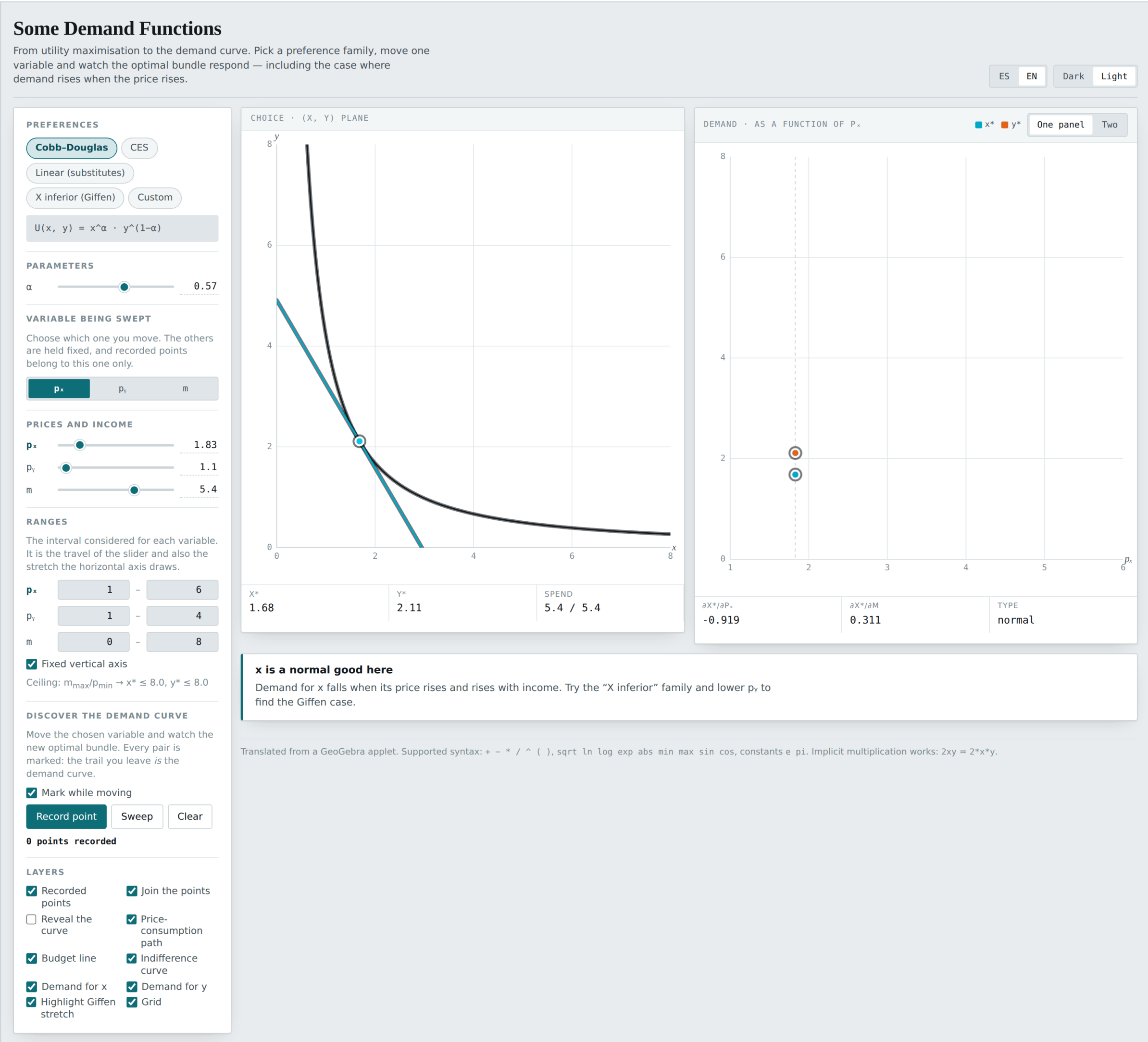
MANUAL

Demand Functions

Where demand curves come from. Move a price or income, and every optimal bundle leaves a mark: the trail you leave is the demand function.



Applet manual · Introductory Microeconomics · Universidad de los Andes
Translated from a GeoGebra applet.

Contents

1 What this is

A demand curve is not a datum. It is the result of solving the consumer's problem once for every price, and writing the answers down.

In a course that gets said, the finished curve gets drawn, and the class moves on. This applet does the other thing: the consumer's problem on the left, an empty pair of axes on the right, and the student moving the price. Every optimal bundle leaves a mark in the right-hand panel. After a while there is a curve there, and the student built it point by point.

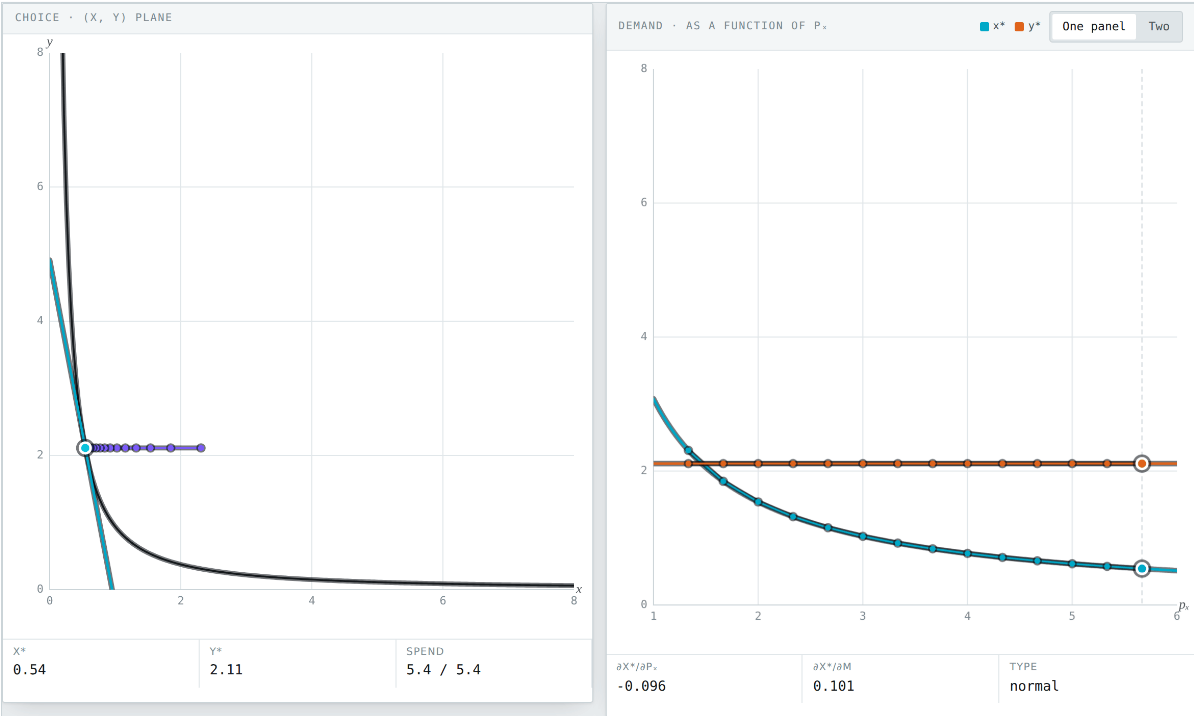


Figure 1. After sweeping p_x. The purple points in the left panel are the price-consumption path; those on the right are the demand curve just built, with the exact curve revealed underneath.

> **The point**
>
> The trail and the curve are not two things: the revealed curve passes exactly through the recorded points. Seeing them coincide is the difference between knowing the definition of a demand function and understanding it.

1.1 Who it is for

The demand topic, after the consumer's optimum has been covered. It is the bridge between the two. It also handles normal, inferior and Giffen goods, which usually survive as a footnote and here can be manufactured and looked at.

2 Opening it

There is nothing to install. The whole application is one `index.html` file: it downloads no libraries, calls no server, and stores nothing outside your browser.

2.1 The three ways

Double-click the file. The quickest route. Find `demand-functions/index.html` and open it. The browser shows it straight off the disk, with an address beginning `file://`. It works offline from the first moment.

From the web. If someone has published it, the address ends in `/demand-functions/`. Once the page has loaded you can disconnect: it needs the network for nothing else.

Served from a folder. For a class, putting the folder on any static server is enough. With Python installed:

```
python3 -m http.server 8000
```

then `http://localhost:8000/demand-functions/` in the browser.

> **Note**
>
> Saving the page with `Ctrl` + `S` (or `Cmd` + `S` on a Mac) leaves a copy of your own that still works offline and that can travel on a memory stick or go out by email.

2.2 Language and theme

Two pairs of buttons sit at the top right.

Control	What it does
ES / EN	Switches the whole interface between Spanish and English on the spot. Nothing is recomputed, so you can switch mid-explanation without losing the state you had.
Dark / Light	Switches the theme. Dark is the original design; light exists for projectors and for printing.

Both choices are remembered in the browser, so next time it opens the way you left it.

2.3 Which browser

Any reasonably recent one: Chrome, Edge, Firefox or Safari, on a computer, tablet or phone. On a narrow screen the panels stack one above the other instead of sitting side by side. No account, no permissions, no extensions.

3 The screen

Ranges. The interval considered for each variable. It is both the travel of the slider and the stretch the right panel's horizontal axis draws. It is the most easily forgotten control and the one that solves the most problems.

Preferences. Five families, the last one writable.

Parameters. Those of the chosen family.

Variable being swept. Which of p_x, p_y or m moves. The other two are held fixed, and recorded points belong to that one only.

Prices and income. p_x, p_y, m.

Discover the demand curve. The buttons that record, sweep and clear.

Layers. What gets drawn.

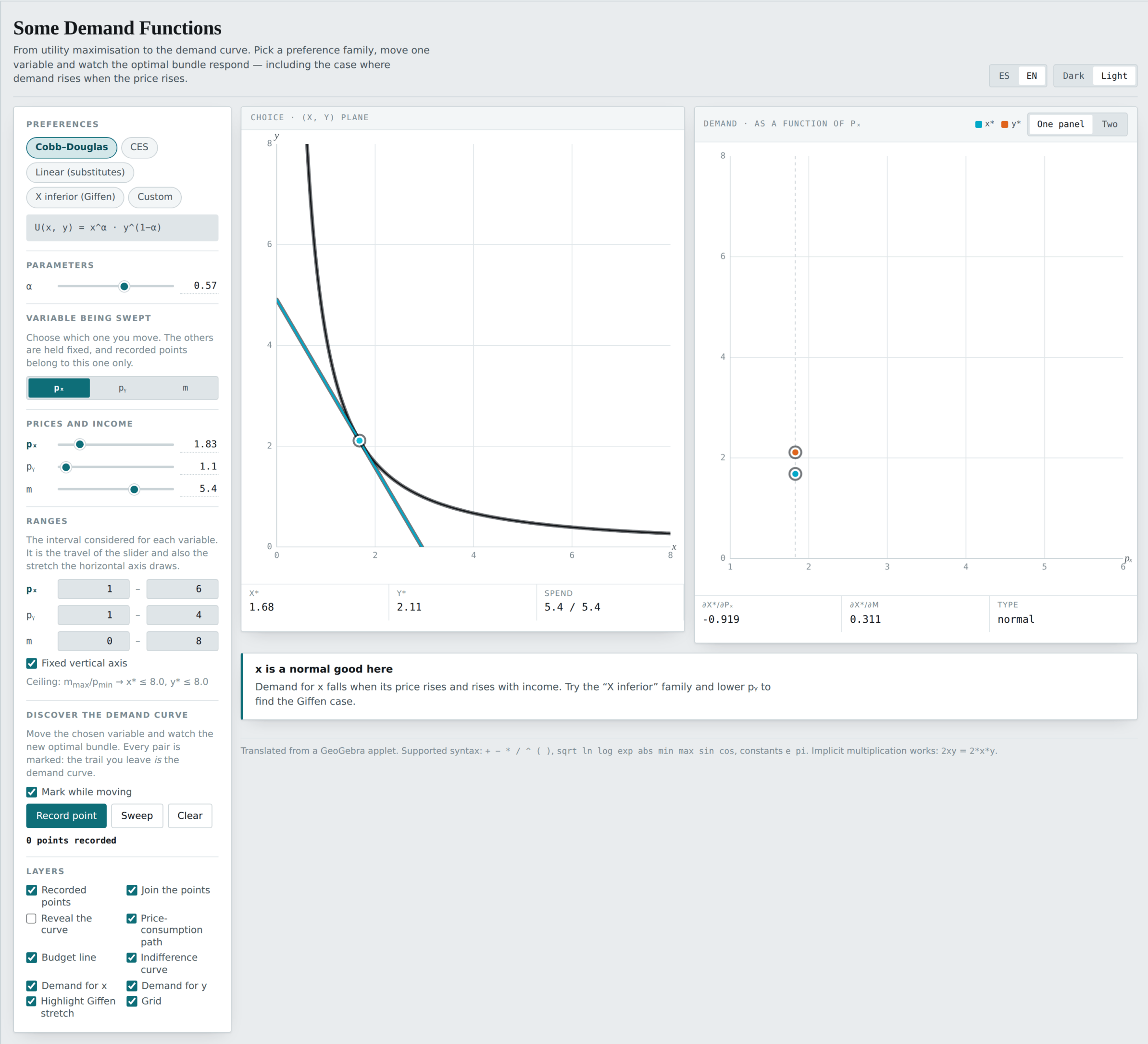


Figure 2. The whole screen. Left: the consumer's problem. Right: the demand curve being built.

3.1 The two panels

Choice · (x, y) **plane.** The budget line, the indifference curve through the optimal bundle, the bundle itself, and the price-consumption path as it forms.

Demand. The optimal quantity against the variable being moved. Its header says which. When the variable is income, the panel becomes the **Engel curve**.

The **One panel / Two** switch in the right-hand header separates the demand for x from that for y into two plots, which is what you need when the two have very different scales.

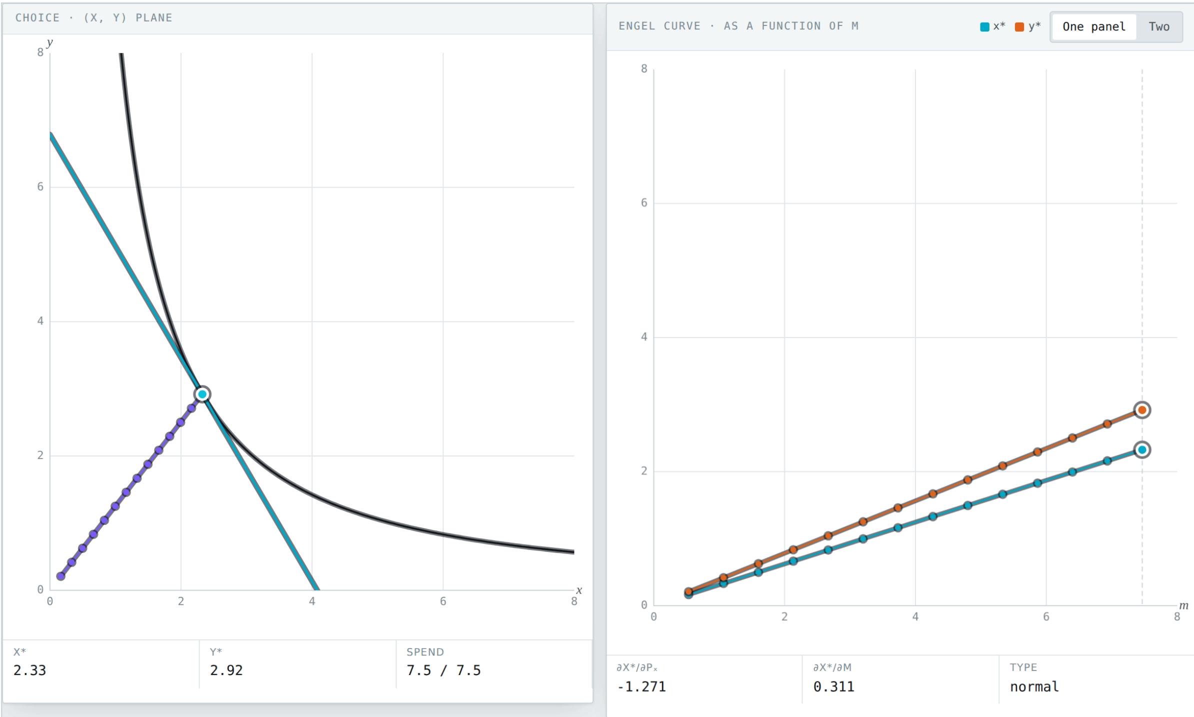


Figure 3. With income as the swept variable, the right panel becomes the Engel curve.

3.2 The dark theme

4 The five families

Each brings a closed-form demand, except the custom one, which is solved numerically.

Family	Utility and demand
Cobb–Douglas	$U = x^{\alpha} y^{1-\alpha}$, with $x^* = \alpha m / p_x$ and $y^* = (1-\alpha) m / p_y$. Each good's demand ignores the other good's price: the clean case.
CES	$U = (x^r + y^r)^{1/r}$. With r near 1 they are near-perfect substitutes; with r strongly negative, near-complements.
Linear (substitutes)	$U = ax + y$. Demand is at a corner: all x if $a > p_x/p_y$, all y if not.
X inferior (Giffen)	$U = \ln(x - \underline{x}) - \frac{1+b}{b} \ln(\bar{y} - y)$ with $b = (1-\beta)/\beta$. This is the family the Giffen case is built with.
Custom	Whatever $U(x, y)$ you type.

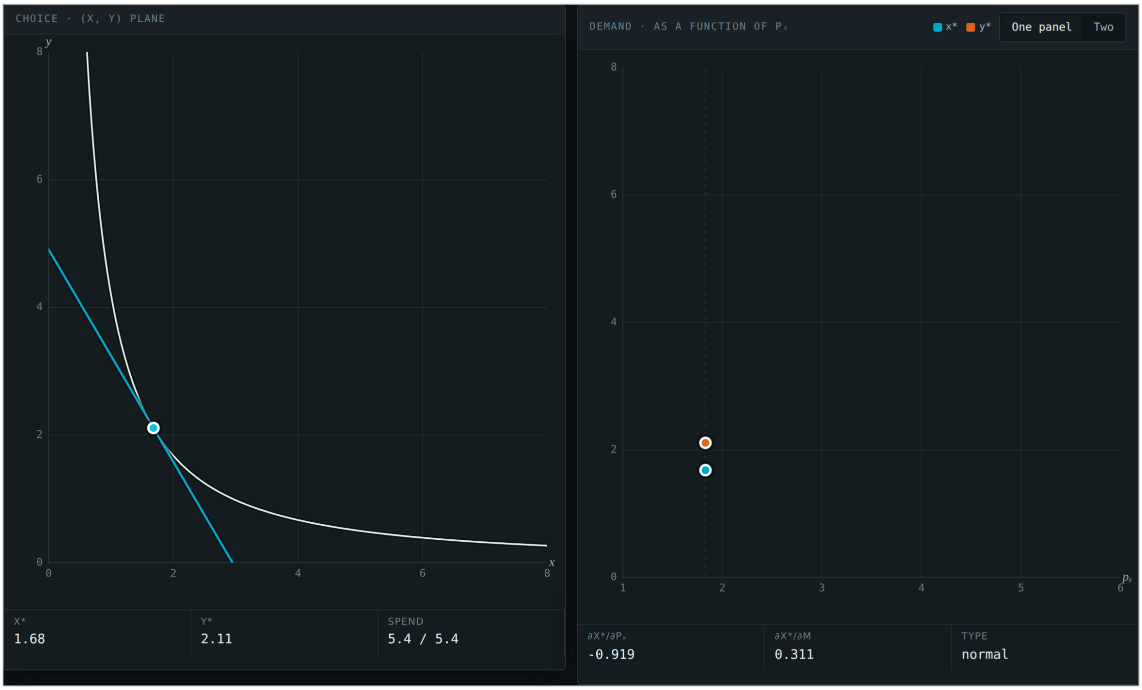


Figure 4. Both panels in the dark theme.

Parameter	Which family
α	Cobb–Douglas: x's share of spending.
r	CES: the substitution exponent.
a	Linear: what x is worth against y.
$\beta, \underline{x}, \bar{y}$	X inferior: the weight, the subsistence minimum of x, and the ceiling on y.

5 Discovering the demand curve

This is the heart of the applet. Four controls:

Control	What it does
Mark while moving	Records a point automatically every time the chosen variable moves. The most comfortable setting for a class demonstration.
Record point	Saves the current bundle. One point, deliberately.
Sweep	Runs the variable from one end of its interval to the other, marking as it goes. Press again to stop.
Clear	Erases the points for the current variable.

Points belong to the variable that was being moved. Switching from p_x to m does not mix the trails: each keeps its own.

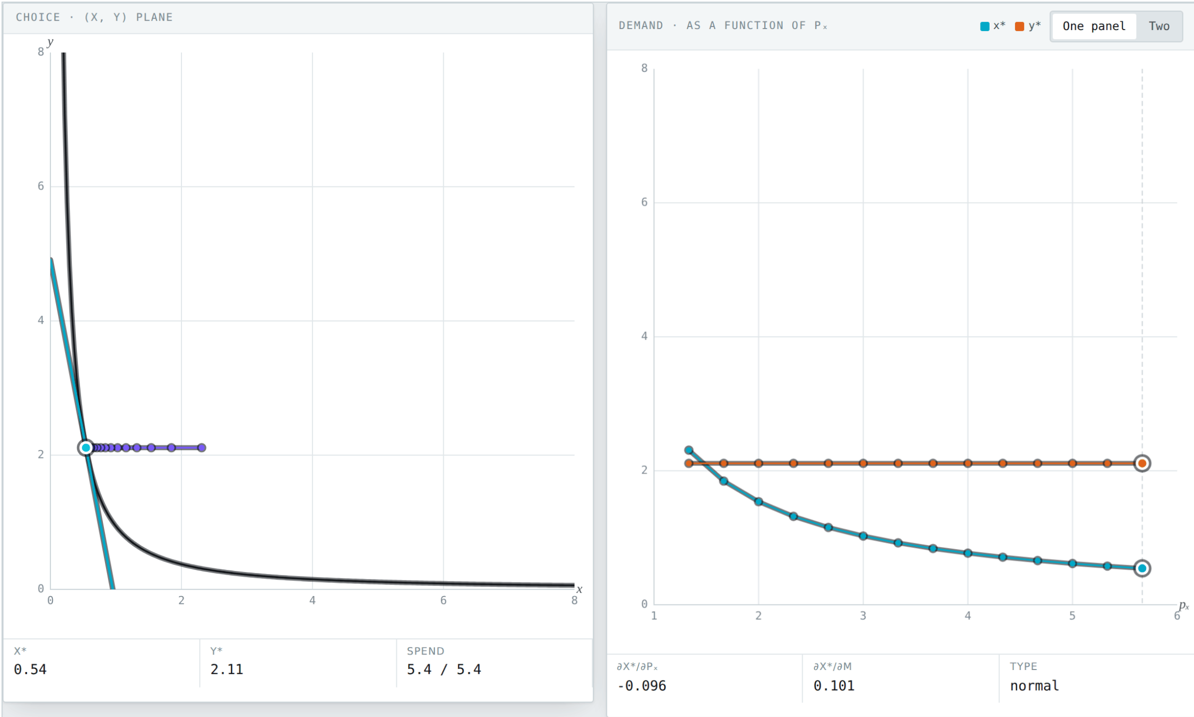


Figure 5. Part-way through a sweep of p_x: the points recorded so far and the current optimal bundle.

5.1 The layers

Layer	What it draws
Recorded points	The marks left so far.
Join the points	Connects them with segments.
Reveal the curve	Draws the exact, computed demand function underneath the points. This is the check.
Price-consumption path	In the (x, y) plane, the trail of optimal bundles.
Budget line	The current constraint.
Indifference curve	The one through the optimal bundle.
Demand for x / Demand for y	Which of the two demands the right panel draws.
Highlight Giffen stretch	Marks in purple the stretches with $\partial x^*/\partial p_x > 0$.
Grid	The background rule.

6 Ranges and axes

> **Watch out**
>
> The p_x, p_y and m sliders are *bounded by the interval* set in the **Ranges** group. If you try to take p_y below its range minimum it will not move. Lower the range minimum first. This is the most frequent cause of a slider that appears not to respond.

The right panel's vertical axis has two modes:

Fixed vertical axis (the default). The ceiling is $m_{\max}/p_{\min}$, the most that can be bought under the current ranges. The advantage is that the axis does not move as you drag, so two sweeps can be compared by eye.

Automatic. The axis fits the curve. You need this when the quantity of interest is small next to that ceiling — the Giffen case, for instance, where x^* hovers around 0.3 and the ceiling is 8.

7 Normal, inferior, Giffen

The three readouts on the right settle everything:

Readout	What it says
$\partial x^*/\partial p_x$	The slope of demand. Negative is ordinary; positive is Giffen.
$\partial x^*/\partial m$	The income effect. Negative means an inferior good.
Type	*normal* or *inferior*, from the sign of the previous one.

And the verdict at the top names the case:

x is a normal good here. Demand falls when its price rises and rises with income.
x is an inferior good here. $\partial x^*/\partial m < 0$. Being inferior is *necessary* for Giffen, but not sufficient.
x is a Giffen good here. $\partial x^*/\partial p_x > 0$. The income effect dominates the substitution effect, which can only happen if x is inferior.

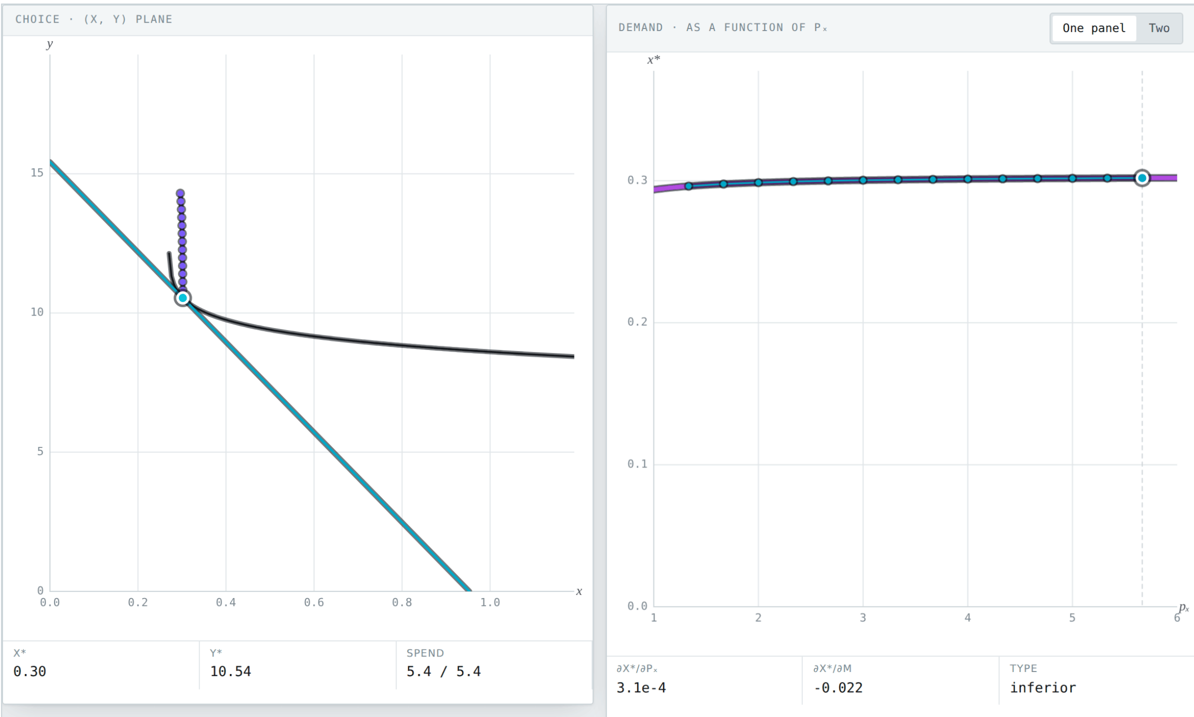


Figure 6. The Giffen case. Demand rises with the price, and the whole stretch is highlighted. Look at the left panel: the price-consumption path climbs almost vertically, because the consumer can barely cut x below their subsistence minimum.

> **x is a Giffen good here**
> Demand for x rises when its own price rises: $\partial x^*/\partial p_x > 0$. The income effect dominates the substitution effect, which can only happen if x is inferior. The upward-sloping stretch is highlighted on the curve.

Figure 7. The verdict in the Giffen case.

8 Classroom activities

Activity 1 · Building a demand curve by hand

Settings: Opening values (Cobb–Douglas). **Variable being swept** on p_x. Switch on **Mark while moving**.

What you should see: At $p_x = 1.83$: $x^* = 1.68$, $y^* = 2.11$, spend 5.4 of 5.4. Move p_x slowly from one end to the other. Every bundle leaves a mark on the right.

Once you have fifteen or twenty points, switch on **Reveal the curve**. The computed curve passes exactly through them. Ask the class what a demand function is, then: not a datum, but the record of twenty optimisation problems solved.

Activity 2 · Spending does not move

Settings: Cobb–Douglas, sweeping p_x.

What you should see: The **Spend** readout always says 5.4/5.4, and y^* does not change as p_x moves. Under Cobb–Douglas each good takes a fixed share of income: $x^* = \alpha m/p_x$ does not involve p_y, and y^* does not involve p_x.

Move α and watch the two shares change. It is the property that makes Cobb–Douglas the exam case, and it is worth seeing before it is assumed.

Activity 3 · The Engel curve

Settings: **Variable being swept** on m. Sweep.

What you should see: The right panel is renamed **Engel curve**. Under Cobb–Douglas it comes out a straight line through the origin: demand is proportional to income.

Switch to **X inferior** and repeat. The Engel curve bends downwards: more income, less x. There is the definition of an inferior good, drawn.

Activity 4 · Inferior is not Giffen

Settings: **X inferior** family, opening values.

What you should see: $x^* = 1.08$, $y^* = 3.12$, $\partial x^*/\partial p_x = -0.422$ and $\partial x^*/\partial m = -0.068$. The type is *inferior* and the verdict says so, but demand still slopes downwards.

Ask whether that contradicts anything. It does not: being inferior is necessary for Giffen, not sufficient. The income effect also has to be strong enough, and for that the consumer has to be made poor in real terms.

Activity 5 · Manufacturing a Giffen good

Settings: **X inferior** family. In **Ranges**, lower the minimum of p_y to 0.2. Then set $p_y = 0.35$. Turn off the **Fixed vertical axis**.

What you should see: Now $x^* = 0.30$, $y^* = 10.54$, and $\partial x^*/\partial p_x = +3.1 \times 10^{-4}$: *positive*. The verdict says “x is a Giffen good here” and the stretch is highlighted.

Sweep p_x and watch the curve rise. What you have done is make the other good so cheap that x has become the consumer’s subsistence good: when x gets dearer they get poorer, and getting poorer they buy more of the cheap staple. It is the Irish potato story, and it takes this much setting up to see it.

Activity 6 · The two extreme cases

Settings: **Linear (substitutes)** family. Sweep p_x.

What you should see: Demand is a step: all x until p_x/p_y crosses a, then zero at once. There is no stretch in between.

Switch to **CES** and take r down towards -6: the curves smooth out until they resemble complements. The two extremes bracket what a demand curve can do, and the other families fall between them.

Activity 7 · The price-consumption path

Settings: Any family. Switch on **Price-consumption path** and sweep p_x.

What you should see: In the (x, y) plane the trail of optimal bundles appears. Under Cobb–Douglas it is horizontal, because y^* does not depend on p_x. Under the inferior family it climbs almost vertically.

The demand curve in the right panel is that same path, with p_x on the horizontal axis instead of x. Same information, two coordinate systems.

Activity 8 · Your own utility function

Settings: **Custom** family. Type `x^0.3*y^0.7`, then `min(2x,y)`, then `x+4ln(y)`.

What you should see: Demand is solved numerically and the trail builds the same way. With `min(2x,y)` the two demands move together; with the quasilinear one, the demand for y stops depending on income once income passes a threshold.

This is the moment to have the class predict the shape before sweeping.

9 Expression syntax

Element	What is allowed
Operators	`+ - * / ^ ( )`
Functions	`sqrt ln log exp abs min max sin cos`
Constants	`e`, `pi`
Implicit multiplication	`2xy` is `2*x*y`
Variables	Only `x` and `y`

10 How it is computed

The four named families carry closed-form demands, derived by hand. The custom one cannot, so its optimum is found numerically along the budget line: a dense scan and a golden-section refinement, the same as in the other applets in this collection. It is robust where a first-order condition finds nothing — corners and kinks included.

The derivatives $\partial x^*/\partial p_x$ and $\partial x^*/\partial m$ in the readouts are central differences on the demand function itself, not symbolic derivatives: that way they work identically for the closed-form families and for whatever the user types.

11 Troubleshooting

Symptom	What is happening
A price slider will not move	It is hitting its range. Change the minimum or maximum under **Ranges**.
The curve is flat against the axis	The vertical axis is fixed at $m_{\max}/p_{\min}$ and the quantity is small. Turn off **Fixed vertical axis**.
The points vanish when I change variable	They do not: each variable keeps its own trail. Go back and they are there.
The right panel is unreadable	The two demands have very different scales. Switch to **Two** panels, or turn one of the layers off.
I cannot get the Giffen case	It takes all three: the **X inferior** family, a low p_y (having lowered its range first), and reading the sign of $\partial x^*/\partial p_x$, not of $\partial x^*/\partial m$.

12 Publishing it for a class

Any static host will do: GitHub Pages, Netlify, a university web directory, even a shared folder served over HTTP. The only thing to upload is the `index.html` file, inside a folder named however you want the address to read.

The repository ships a GitHub Actions workflow (`.github/workflows/pages.yml`) that publishes every applet at once on each push to the default branch. To switch it on, in the repository: *Settings* → *Pages* → *Source: GitHub Actions*. That is the one step the workflow cannot do for itself, because creating a Pages site needs administration rights that a `GITHUB_TOKEN` is never granted.

> **Note**
>
> To hand the applet to students without reliable internet, send them the `index.html` file directly. It weighs less than a photograph and opens with a double-click.

13 Licence and provenance

The application code is MIT. Use it, change it and pass it on, in class or out of it, with attribution.

This applet is a standalone rewrite of an original GeoGebra applet. The rewrite is not a literal translation: where the original had a construction error, a division by zero reachable with its own sliders, or a dead object inherited from whatever applet it was copied from, this version fixes it and says so. The section *Changes from the GeoGebra original* lists those changes one by one.

Everything is hand-written and dependency-free: the expression parser, the symbolic differentiation, the level-curve tracing and the drawing. In particular there is no `eval`, so what a student types into a text box cannot become code, and a strict content security policy does not break the page.